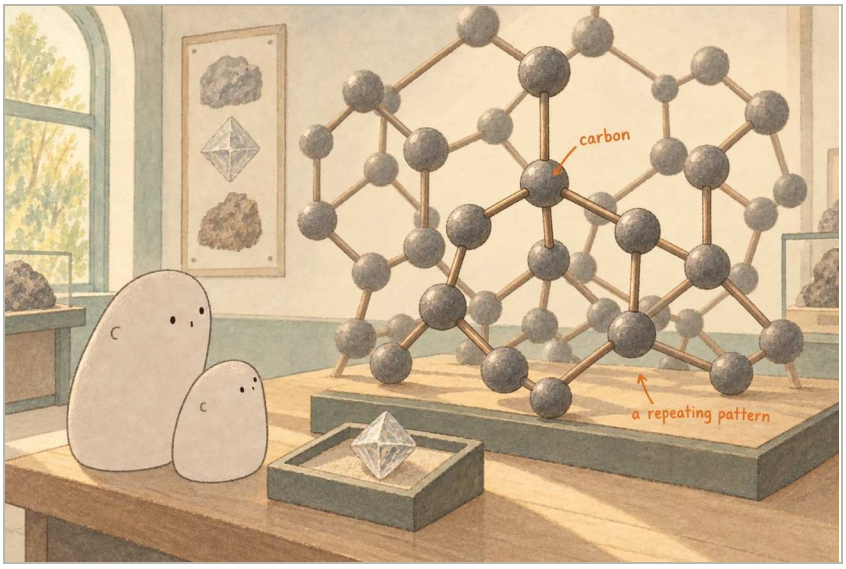
Parts for a pattern

The flash that wouldn't stay. · Curiosity day

Name _____ Date _____



The model reveals a repeating carbon pattern—not a hidden source of light. Real atoms are much too small to see.

PRETEND MODEL SUPPLIES · RODS

Tray	A	B
Rods	18	7

Use these invented supply counts. Do not count the rods in the painting.

1. Tray A has 18 rods. Tray B has 7. How many rods altogether?

18 + 7 = _____

Show your thinking.

2. You have 25 rods. Set 9 aside for later. How many can you use now?

25 − 9 = _____

Show your thinking.

3. Draw 3 separate groups with 4 rods in each. How many rods did you draw?

4 + 4 + 4 = _____

Show your thinking.

These are invented supply counts, not a count of atoms in a diamond. Try drawing a different way to group 12 rods.

No sun down here

A flash winks from a museum case. "The diamond makes light!" says Little Wonderer. Big points to its deep beginning.

A pattern, not a lantern

"Maybe light hides in the tiny parts," Little says. Balls and rods stand for carbon atoms and their connections.

The long way to daylight

Little traces an arrow up. "To meet light, it had to leave the dark!"

The strong-stone surprise

A label marked 10 stops Little. Diamond tops the Mohs scratch scale. "So nothing can hurt it?"

A borrowed beam

At the last model, a lamp clicks on. "There!" Little cries. Light enters the diamond, bends, bounces, and comes out again.

The flash that won't stay

Little shifts. One bright face dims; another flashes. "Found you!" The flash was not stored in the stone.

Make a little book

1 Fold it into 8 boxes. Open it flat.

2 Fold short end to short end. Cut the thick line. Open it.

3 Fold the long way, pictures out. Push the ends in.

4 Fold into a book. Page 1 goes on the front!

WONDER DAILY

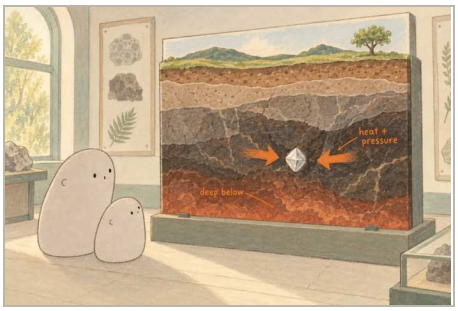
The flash that wouldn't stay.

Keep a clue from the chase.

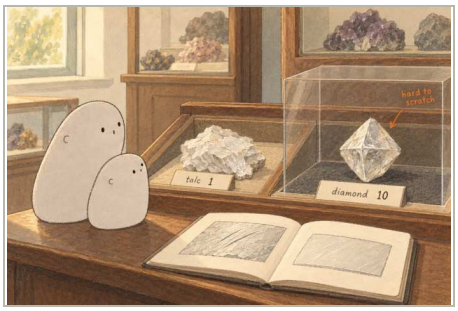
The flash that wouldn't stay. · Curiosity day

Name _____ Date _____

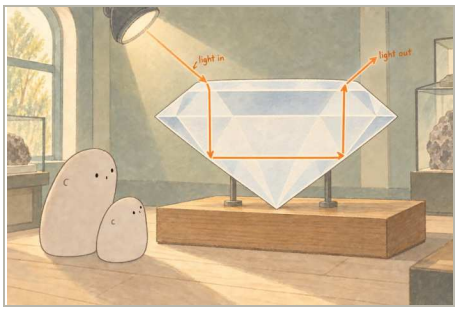
Choose a picture from Little's search for the flash. Say what changed Little's mind. Then draw or write one idea you want to keep.



No sun down here



The strong-stone surprise



Where the flash begins

Draw one detail from your scene. Add an arrow or a label to help someone understand it.

carbon · crystal · heat · pressure · scratch · light · facet

Deep underground, … What helped the diamond crystal grow?

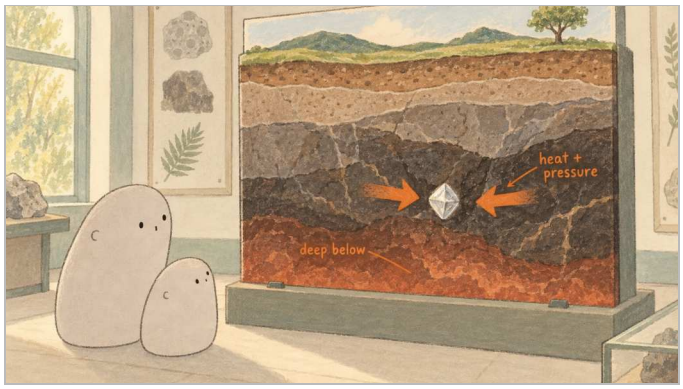
A diamond is hard to scratch, but … What could still happen to it?

The flash begins when … What gives the light? How do the facets and watching eyes help?

Read it together. Point to the detail your words explain. If you are writing, check the first capital and the ending mark.

The flash that wouldn't stay.

01 Could a diamond flash where no light can reach it?



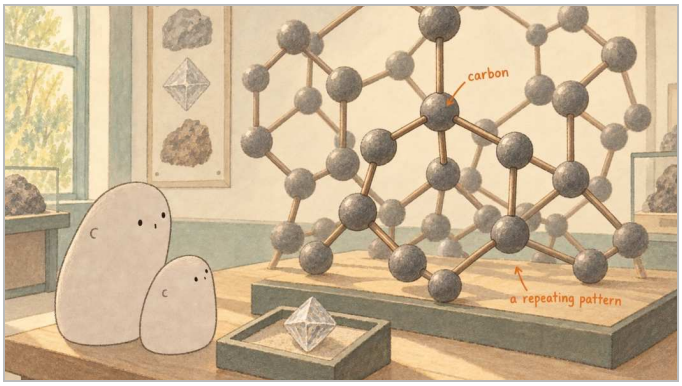
An imagined window into the diamond's dark beginning. The crystal is enlarged; layers and distances are simplified.

Look closer. Find the tiny landscape, the crystal, and the two orange arrows. What looks bright? What looks dark?

A little explanation. Natural diamonds grew from carbon deep underground, where it was very hot and the pressure was enormous. With no light reaching it, a diamond there could not send a flash to anyone's eyes.

Try this. Cover the crystal in the picture with one finger, then uncover it. Tell your grown-up where you would search next for the flash.

02 Could the diamond's tiny pattern be making the flash?



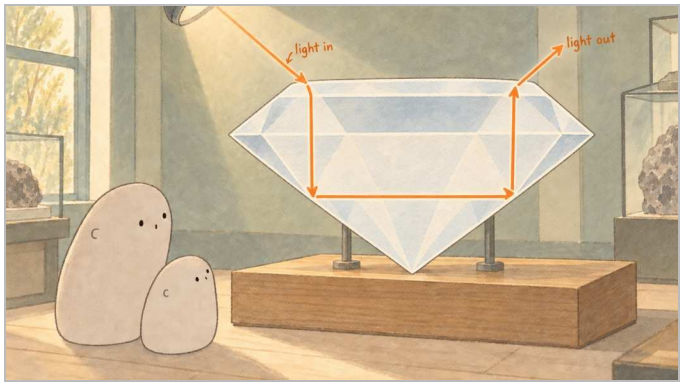
The model reveals a repeating carbon pattern—not a hidden source of light. Real atoms are much too small to see.

Look closer. Look from the small stone to the large model. Follow one rod to the next. Do you see a lamp or another source of light?

A little explanation. A diamond is made mostly of carbon atoms joined in a repeating pattern. The balls and rods help us picture parts far too small to see, but the pattern is not a hidden lamp.

Try this. Trace three connections with your finger. Say what this model explains—and what it still does not explain.

03 Which thing in this picture starts the bright path?



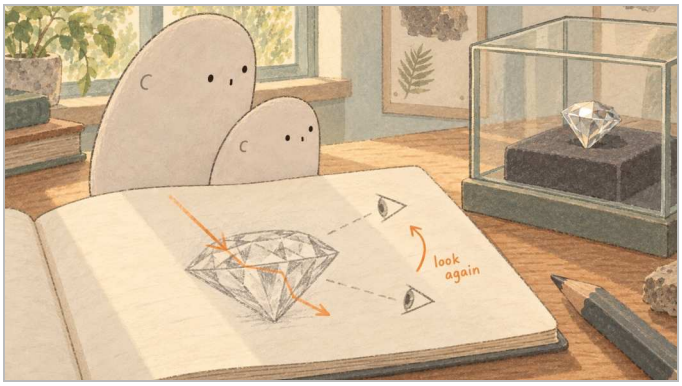
The lamp begins this simplified path. Cut facets change the light's direction and send some of it back out.

Look closer. Begin at the orange line. Follow it backward until you cannot go any farther. Where did you land?

A little explanation. The path begins at the lamp. Light bends when it enters the diamond and can bounce inside. Carefully cut facets help send some of that borrowed light back toward our eyes.

Try this. Trace the whole path from lamp to diamond to eye. Imagine the lamp switched off. What part of the path would disappear?

04 Why might the flash move when Little moves?



Little's two viewing spots turn a quick flash into an observation they can keep.

Look closer. Find the two eye positions in the sketch. Follow each line toward the drawn diamond. What changed?

A little explanation. A flash reaches our eyes only along certain paths. When the light, diamond, or viewer moves, a bright facet can dim while another one flashes.

Try this. Point to one eye position, then the other. Pretend you are Little and tell where you would look next.